Jie Song

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Education

visiting Ph.D. program — University of Tokyo, *May 2025 - May 2026*
- **Major**: Astrophysics
- **Supervisor**: Prof. Masami Ouchi

Ph.D. — University of Science and Technology of China, *Sep 2023 - Oct 2026*
- **Major**: Astrophysics
- **Supervisor**: Prof. Xu Kong

M.D. — University of Science and Technology of China, *Sep 2021 - Jul 2023*
- **Major**: Astrophysics
- **Supervisor**: Prof. Xu Kong

B.D. — University of Science and Technology of China, *Sep 2017 - Jul 2021*
- **Major**: Theoretical Physics
- **Supervisor**: Prof. Xu Kong

Project Involved

CSST Galaxy Science Research Project
- Software development for galaxy nonparametric morphology measurement
- Software development for galaxy morphology classification using Machine Learning
- Analyze the effect of environment on galaxy properties at high redshift

CSST IFS instrument performance evaluation
- CSST IFS data analysis pipeline development
- Evaluate the performance of CSST IFS in recovering galaxy star formation histories

Conference Attendance

2023 Workshop on Key Issues in CSST Galaxy Science Research, oral — **2023**
Workshop on Galaxy Observation and Theory in the Space Telescope Era, oral — **2023**
Workshop on the Morphology and Structure of high Redshift galaxies, oral — **2024**
2024 Workshop on Key Issues in CSST Galaxy Science Research, oral — **2024**

Tools & Software

- **Image Analysis**: SExtractor, photutils, GALFIT, GALFITM, statmorph, USmorph
- **SED Analysis**: FAST, CIGALE, BAGPIPES, fsps, pPXF, PHANGS-MUSE DAP, grizli
- **Software Development**: statmorph_csst

Awards

First-prize Academic Scholarship (Four times) — **2021 - 2024**
Second-prize Academic Scholarship — **2025**
The Funding of Overseas Study for Excellent PhD Students in USTC — **2025**

Teaching experience

Mechanics — **2024**
Thermal physics — **2024**

Publication List

First-Author

- **Song, J.**, Fang G., Gu Y., Lin Z., and Kong X. (2023). The Effect of Environment on the Properties of The Most Massive Galaxies at $0.5 < z < 2.5$ in the COSMOS-DASH Field. ApJ, 950, 130 10.3847/1538-4357/acd174
- **Song, J.**, Fang G., Lin Z., Gu Y., and Kong X. (2023). Solution to the Conflict Between the Estimation of Resolved and Unresolved Galaxy Stellar Mass from the Perspective of JWST. ApJ, 958, 82 10.3847/1538-4357/ad0365
- **Song, J.**, Fang G., Ba S., Lin Z., Gu Y., et al. (2024). USmorph: An Updated Framework of Automatic Classification of Galaxy Morphologies and Its Application to Galaxies in the COSMOS Field. ApJS, 274, 42 10.3847/1538-4365/ad434f
- **Song, J.**, Wang E., Jia C., Lyu C., Chen Y., Wang J., Li F., Ding W., Fang G., and Kong X. (2025). Transition from Outside-in to Inside-Out at $z \sim 2$: Evidence from Radial Profiles of Specific Star Formation Rate based on JWST/HST. Submitted, https://arxiv.org/abs/2512.01684

Co-Author

- Lyu C., Wang E., Wang J., Jia C., **Song J.**, et al. (2025). Central Concentration and Escape of Ionizing Photons in Galaxies at the Epoch of Reionization. ApJL, 988, L72 10.3847/2041-8213/adf0f9
- Jia C., Wang E., Lyu C., Ma C., **Song J.**, et al. (2025). Potential-driven Metal Cycling: JADES Census of Gas-phase Metallicity for Galaxies at $1 < z < 7$. ApJL, 986, L24 10.3847/2041-8213/addfd9
- Zhu S., Fang G., Zhou C., **Song J.**, Lin Z., et al. (2025). Dual-coding Contrastive Learning Based on the ConvNeXt and ViT Models for Morphological Classification of Galaxies in COSMOS-Web. ApJS, 278, 39 10.3847/1538-4365/add0b8
- Fang G., Dai Y., Lin Z., Zhou C., **Song J.**, et al. (2025). An efficient unsupervised classification model for galaxy morphology: Voting clustering based on coding from ConvNeXt large model. A&A, 693, A141 10.1051/0004-6361/202451734
- Jia C., Wang E., Wang H., Yao Y., **Song J.** et al. (2024). Size Growth on Short Timescales of Star-forming Galaxies: Insights from Size Variation with Rest-frame Wavelength with JADES. ApJ, 977, 165 10.3847/1538-4357/ad919a
- Zhang S., Fang G., **Song J.**, Li R., Gu Y., et al. (2024). Preparation for CSST: Star-galaxy Classification using a Rotationally Invariant Supervised Machine Learning Method. RAA, 24, 095012 10.1088/1674-4527/ad6fe6
- Yao Y., **Song J.**, Kong X., Fang G., Zhang H., et al. (2023). Evolution of Nonparametric Morphology of Galaxies in the JWST CEERS Field at $z \simeq 0.8 \sim 3.0$. ApJ, 954, 113 10.3847/1538-4357/ace7b5
- Dai Y., Xu J., **Song J.**, Fang G., Zhou C., et al. (2023). The Classification of Galaxy Morphology in the H-Band of the COSMOS-DASH Field: A Combination-based Machine-learning Clustering Model. ApJS, 268, 34 10.3847/1538-4365/ace69e
- Ding W., Zou H., Kong X., Gao Y., ..., **Song J.**, et al. (2023). Strong $[OIII]$ $\lambda 5007$ Compact Galaxies Identified from SDSS DR16 and Their Scaling Relations. ApJ, 166, 133 10.3847/1538-3881/ace893
- Fang G., Ba S., Gu Y., Lin Z., ..., **Song J.**, et al. (2023). Automatic Classification of Galaxy Morphology: A Rotationally-invariant Supervised Machine-learning Method Based on the Unsupervised Machine-learning Data Set. AJ, 165, 35 10.3847/1538-3881/aca1a6